

In class exercises: Big-O and Big-Ω

1. Sort the following functions from asymptotically smallest to asymptotically largest, indicating ties if there are any. You do not need to do formal proofs, but it's a good idea to make sure you understand how you would prove these and what facts/identities are needed to simplify.

1	n	$3n^2 - 6n + 12$	$\lg n$	$\lg \sqrt{n}$
$\cos n + 2$	$n^{\lg n}$	$\lg 2^n$	$\sum_{i=1}^n \sum_{j=1}^i 1$	$2^{2 \lg n}$
$\sqrt{2^{\lg n}}$	$n!$	$n \lg n$	$\sum_{i=1}^n i^2$	$\lg(\sqrt{2^n})$

To simplify notation, write $f(n) \ll g(n)$ to mean $f(n) = o(g(n))$ and $f(n) \equiv g(n)$ to mean $f(n) = \Theta(g(n))$. For example, the functions n^2 , n , $\binom{n}{2}$, n^3 could be sorted either as $n \ll n^2 \equiv \binom{n}{2} \ll n^3$ or as $n \ll \binom{n}{2} \equiv n^2 \ll n^3$. [Hint: When considering two functions $f(\cdot)$ and $g(\cdot)$ it is sometime useful to consider the functions $\ln f(\cdot)$ and $\ln g(\cdot)$.]

Hint: You may want to remember some summation and logarithm identities. A handy (if overwhelming!) guide is here: <https://tug.org/texshowcase/cheat.pdf>.